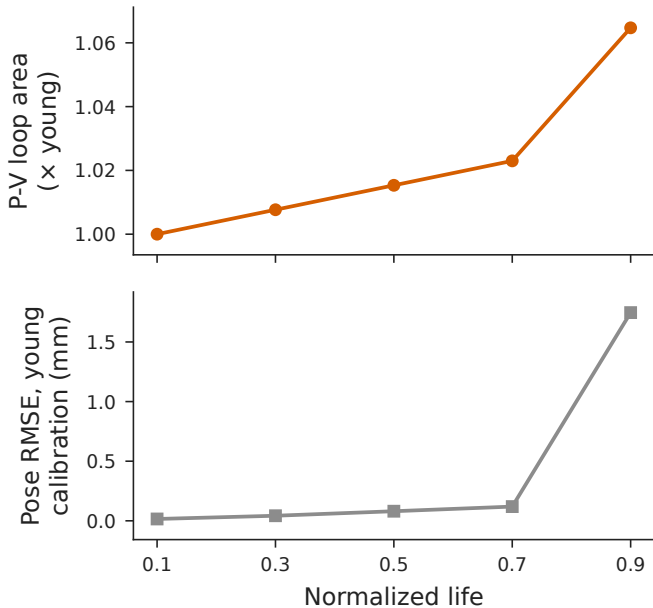


P-V loop area and young-calibration pose error both rise with life



Simulation: Phase D cohort, mean of 6 held-out actuators. Pooled $r = 0.885$ over actuator $\times$ stage points. One fatigue law drives both signals, so the association is built in (`matched_clock_audit.json`).